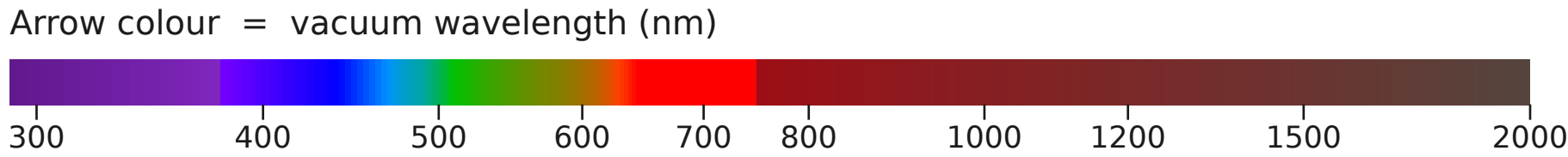


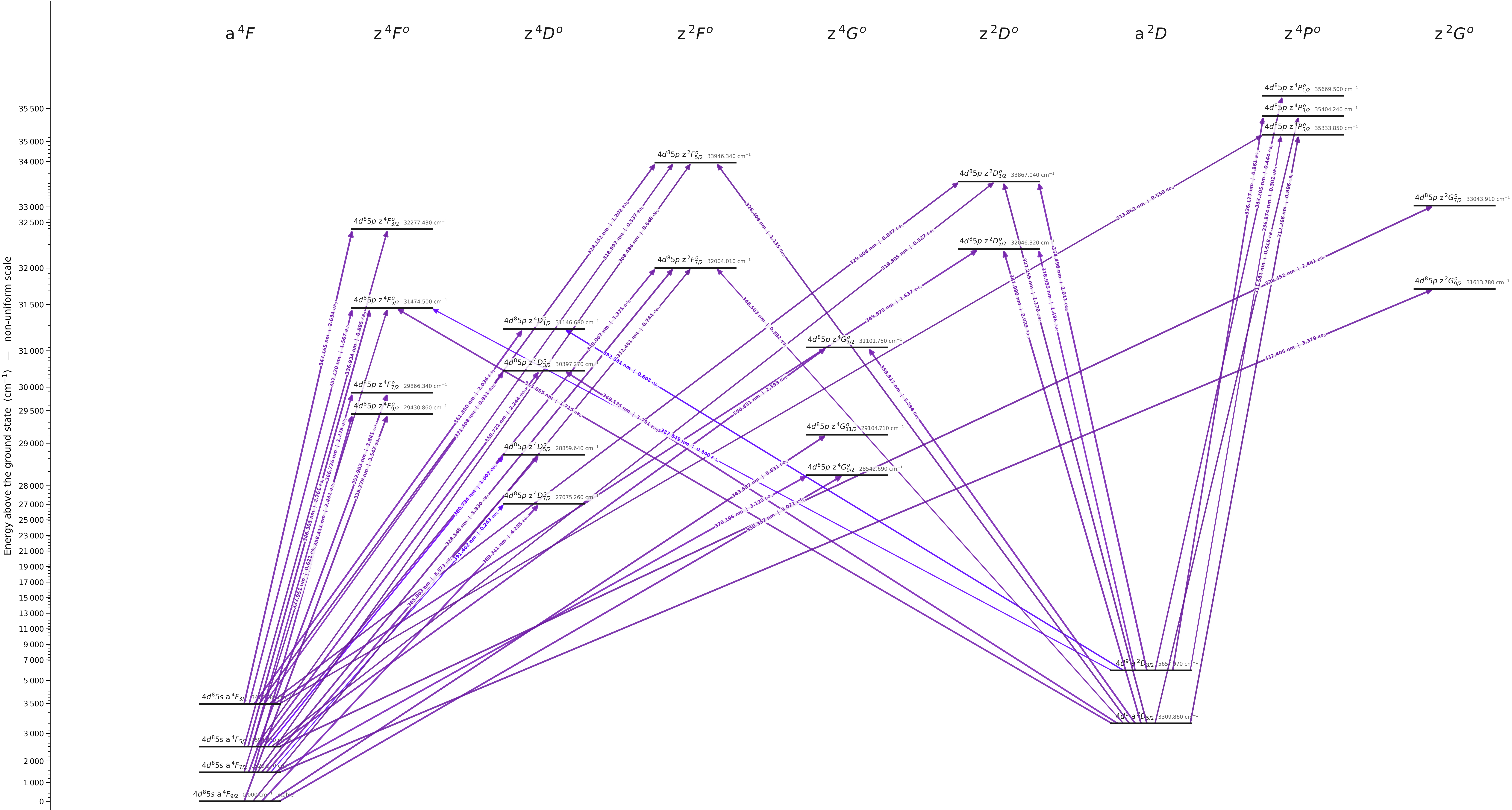
Rhodium

Arrow label: vacuum wavelength | reduced dipole matrix element $[U\|er\|J']$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Rh (Rh I). Strongest classified transitions below 2 μm (NIST): 26 levels, 48 lines.

Rh⁺ ionisation limit 60160.10 cm^{-1} = 7.45890 eV (166.223 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$[U\ er\ J']$ (ea_0)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4d^8 5s a^4 F_{5/2} - 4d^8 5p z^2 G_{7/2}^o$	328.4517	328.3571	912.7445	2.481	4.400e+07	-	-	-
$4d^8 5s a^4 F_{7/2} - 4d^8 5p z^2 G_{9/2}^o$	332.4047	332.3091	901.8899	3.379	6.300e+07	-	-	-
$4d^8 5s a^4 F_{9/2} - 4d^8 5p z^4 F_{9/2}^o$	339.7794	339.6819	882.3150	3.547	6.500e+07	-	-	-
$4d^8 5s a^4 F_{9/2} - 4d^8 5p z^4 G_{11/2}^o$	343.5870	343.4885	872.5373	5.631	1.320e+08	-	-	-
$4d^8 5s a^4 F_{5/2} - 4d^8 5p z^4 F_{9/2}^o$	346.3027	346.2036	865.6948	2.761	6.200e+07	-	-	-
$4d^8 5s a^4 F_{3/2} - 4d^8 5p z^4 F_{9/2}^o$	347.1650	347.0656	863.5447	2.634	8.400e+07	-	-	-
$4d^8 5s a^4 F_{9/2} - 4d^8 5p z^4 G_{9/2}^o$	350.3524	350.2522	855.6883	3.021	4.300e+07	-	-	-
$4d^8 5s a^4 F_{5/2} - 4d^8 5p z^4 G_{7/2}^o$	350.8314	350.7311	854.5200	2.393	3.360e+07	-	-	-
$4d^8 5s a^4 F_{7/2} - 4d^8 5p z^4 F_{9/2}^o$	352.9034	352.8025	849.5030	3.841	8.500e+07	-	-	-
$4d^9 a^2 D_{3/2} - 4d^8 5p z^2 D_{3/2}^o$	354.4959	354.3946	845.6866	2.011	4.600e+07	-	-	-
$4d^8 5s a^4 F_{5/2} - 4d^8 5p z^4 D_{3/2}^o$	359.7221	359.6194	833.4003	2.244	5.480e+07	-	-	-
$4d^9 a^2 D_{5/2} - 4d^8 5p z^4 G_{7/2}^o$	359.8172	359.7146	833.1799	3.294	5.900e+07	-	-	-
$4d^8 5s a^4 F_{3/2} - 4d^8 5p z^4 D_{1/2}^o$	361.3500	361.2470	829.6457	2.036	8.900e+07	-	-	-
$4d^8 5s a^4 F_{7/2} - 4d^8 5p z^4 D_{5/2}^o$	365.9027	365.7985	819.3229	3.573	8.800e+07	-	-	-
$4d^8 5s a^4 F_{9/2} - 4d^8 5p z^4 D_{7/2}^o$	369.3409	369.2358	811.6959	4.255	9.100e+07	-	-	-
$4d^8 5s a^4 F_{7/2} - 4d^8 5p z^4 G_{9/2}^o$	370.1960	370.0906	809.8210	3.125	3.900e+07	-	-	-

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.